

Digital Twin Monitoring System for Pedestrian Crossings Based on Autonomous Driving Environment

Xiao TENG^a, Lin HUANG^a, Zhenjiang SHEN^{b,1} and Wankai LI^b

^a *Graduate School of Natural Science & Technology, Kanazawa University, Kanazawa, Japan*

^b *International Joint Laboratory of Spatial Planning and Sustainable Development (FZUKU-LAB SPSPD), Fuzhou University, Fuzhou, 350025, China*

Abstract. In response to the growing demand for pedestrian crosswalk safety and visibility in the era of autonomous driving, this paper proposes and designs a crosswalk environment monitoring system based on virtual-physical twin technology. Leveraging Unreal Engine (UE) and Carla simulation software, the system constructs a high-precision virtual environment, integrating real-time data from on-site sensors and weather APIs to achieve comprehensive and dynamic monitoring of crosswalk conditions. A key innovation of the system is the application of a Vision-Language Model (VLM), which automatically translates monitoring data into understandable natural language text while also enabling threshold-based warnings and safety alerts. This enhances the intelligence and automation of the monitoring system. Compared to traditional monitoring methods that primarily rely on graphical visualization, the proposed approach significantly improves data processing efficiency and safety alert capabilities, offering a more efficient, real-time, and intelligent solution for traffic safety management in the context of autonomous driving.

Keywords. Carla virtual simulation, system integration, sensor monitoring, visual language large model, environmental monitoring

1. Introduction

With the rapid development of autonomous driving technology, real-time environmental perception and decision-making systems are profoundly reshaping the modern transportation system. As one of the core scenarios of human-vehicle interaction, the safety and recognizability of crosswalks face severe challenges in complex urban environments [1]. Since crosswalks are where pedestrians and vehicles most often come into contact, the probability of accidents involving autonomous vehicles in crosswalks increases significantly. The main reasons include the uncertainty of dynamic pedestrian behavior, perception degradation in extreme weather, and the response lag of traditional monitoring systems [2]. Although existing traffic regulations have clearly defined the physical marking standards for crosswalks, it is still difficult for autonomous vehicles to achieve robust pedestrian intention analysis under multi-

¹ Corresponding Author: Zhenjiang Shen, shenzhe@se.kanazawa-u.ac.jp

source environmental interference (such as heavy rain and low light at night). Therefore, it is urgent to build a new intelligent monitoring paradigm that adapts to the needs of the autonomous driving era.

The current mainstream monitoring technology mainly relies on fixed sensors and visual monitoring platforms. The former collects basic traffic flow data through geomagnetic coils or cameras, but fails to integrate environmental variables such as weather and visibility [3]. Although the latter presents video streams and heat maps in a graphical interface, it still requires manual interpretation of potential risks, resulting in response delays [4]. More importantly, the traditional threshold alarm mechanism cannot effectively identify complex risk scenarios such as "group red light running in heavy rain", exposing systemic defects such as insufficient multimodal data coordination, passive risk warning lag, and semantic barriers in human-computer interaction [5]. In this context, the integration and innovation of digital twin technology and vision-language models (VLMs) [6] provides a new path to solve the above problems. By building a high-fidelity virtual environment of the CARLA simulation platform on the UE, dynamic mapping of physical space and synchronization of multi-source data can be achieved; and the semantic parsing engine driven by VLMs can convert visual data into interpretable natural language descriptions, giving the autonomous driving system human-like environmental cognition capabilities.

This study proposes a digital twin monitoring system for pedestrian crossings in an autonomous driving environment. Its core innovation lies in breaking through the cognitive boundaries of traditional visual monitoring. The system builds a virtual-real fusion perception framework through the UE engine and CARLA, integrates lidar point cloud, meteorological API real-time data, and vehicle perception signals, and establishes a logical mapping table between virtual and real. On this basis, the visual language big model is introduced to realize the "image-text-monitoring" closed-loop analysis, which not only automatically generates a structured risk report, but also provides threshold warnings for the environment based on traffic density and meteorological intensity, thereby improving the safety of road monitoring. Compared with existing systems, this architecture realizes the deep integration of semantic-level environmental understanding and physical simulation for the first time, promoting the paradigm shift of monitoring mode from "passive response" to "active prediction". At the application level, the natural language report output by the system can simultaneously serve the traffic management center and autonomous driving vehicles, effectively reducing the decision-making misjudgment rate by eliminating the semantic gap in human-computer interaction. The article structure of this paper is as follows: Chapter 2 reviews the relevant research progress of digital twins in intelligent transportation; Chapter 3 analyzes the construction of the system theoretical framework; Chapter 4 verifies the system effectiveness through experimental results of simulation experiments; Chapter 5 discusses technical limitations and improvement directions; Chapter 6 summarizes the research results and its generalization potential in smart city transportation construction.

2. Related Work

The core breakthroughs of autonomous driving technology are concentrated in the fields of environmental perception and decision optimization. Vision and LiDAR fusion algorithms represented by YOLOv7 [7] and PointPillars [8] have significantly

improved the target detection accuracy of vehicles on structured roads. However, there is still significant performance degradation in complex urban scenes, especially in crosswalks. Studies have shown that severe weather can cause LiDAR signal scattering, generate measurement noise, and affect scene understanding algorithms. In safety-critical applications, measures must be taken to reduce the degradation of perception performance or identify sensor limitations to ensure prediction reliability [9]. These limitations highlight the urgent need for traditional perception algorithms to fuse multimodal environmental data.

The application of digital twin technology in the transportation field has expanded from infrastructure modeling to dynamic system simulation. The integrated framework of CARLA and SUMO realizes large-scale traffic flow simulation, but existing research focuses on vehicle dynamics modeling, and the fidelity of pedestrian micro-behavior simulation is insufficient [10]. Recent studies have attempted to combine the physics engine of Unreal Engine (UE) with the meteorological particle system to build a visibility attenuation model in rainy and foggy environments, but have not solved the problem of real-time synchronization between virtual sensor data and real IoT devices. It is worth noting that the semantic digital twin architecture proposed by Wang et al. introduces scene graph technology to describe the relationship between road elements, but its textual output is limited to static attribute annotation and cannot parse dynamic risk events at the semantic level [11].

Traditional environmental monitoring relies on fixed sensor networks, and collects parameters such as temperature, humidity, and PM2.5 through low-power wide area networks such as LoRaWAN [12]. In traffic scenarios, Yang et al. proposed a traffic condition recognition method based on millimeter-wave radar, combined with spectral clustering and neural network algorithms, which can adjust the status of traffic lights according to the degree of road congestion and effectively reduce congestion by 20% [13]. Liu proposed a method of "automatically generating lane-level intersection maps using low-channel roadside lidar", focusing on dynamically generating detailed intersection maps based on vehicle trajectories and traffic rules [14]. The spatial resolution of Laner millimeter-wave radar will be attenuated in rainy and foggy weather [15].

Existing traffic management platforms mostly use BIM-GIS fusion visualization technology to present traffic flow heat maps and event alarms through digital signage [16]. Although deep learning technology has achieved automatic detection of traffic accidents, the system response still relies on manual confirmation, and the average handling still has a long time delay [17].

Existing research still has obvious gaps in the following aspects: (1) There is a gap between environmental monitoring and risk warning between different data modalities, and the closed-loop integration of visual information, IoT sensor data and semantic analysis has not yet been achieved; (2) The traffic management decision chain has not yet been able to establish real-time semantic interaction with the autonomous driving system. To overcome these limitations, this paper proposes a VLM-enhanced digital twin architecture that achieves three innovative integrations through digital twin technology: (1) Real-time bidirectional mapping of virtual and real spaces to build a parallel system with narrative integration capabilities; (2) Deep semantic understanding of cross-modal data to establish an end-to-end analysis chain from pixels to decisions; (3) An integrated mechanism of pre-rehearsal and deduction to support the optimization of risk intervention strategies based on early warning.

3. Theoretical Framework Construction

This study uses a method that combines virtual simulation environment construction, real-time data collection and fusion, and intelligent analysis and early warning to establish a digital twin system for pedestrian crossings in the era of autonomous driving, aiming to improve the safety and real-time monitoring level of pedestrian crossings. In terms of virtual simulation environment, this study builds and compiles the CARLA platform [18] based on the UE. Through the simulation platform, the digital twin world is visualized, the monitoring screen is displayed in real time, and the real-time semantic information is presented based on the large language model (VLM). The real-time data collection part obtains dynamic monitoring data of pedestrian crossings by deploying sensor components on site, and uploads it to the system for processing and visualization. The intelligent analysis and early warning module uses the collected data for in-depth analysis, and realizes an automatic alarm mechanism in combination with the set safety threshold, to timely warn of potential risks. The overall research theoretical framework construction path is shown in Figure 1, which fully reflects the organic combination of virtual simulation, real-time data and intelligent analysis, and provides an innovative solution for pedestrian crossing safety monitoring in the era of autonomous driving.

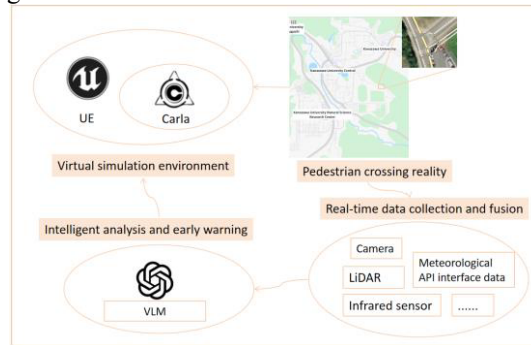


Figure 1. Theoretical framework construction path map

3.1. Virtual Twin Environment System Architecture Design

This study focuses on the architecture design of the virtual twin environment system, following the classic architecture model of the Internet of Things. The overall system consists of four parts: perception layer, network layer, data layer, and application layer. As shown in Figure 2, the operation of the system starts from the perception layer. By deploying cameras, LiDAR, infrared sensors, laser sensors, pressure sensors and other devices around the crosswalk, real-time collection of information such as pedestrians, vehicles, traffic light status, and road environment is achieved. To reduce the impact of data transmission delay on the safety of autonomous driving, this study introduces edge computing in the perception layer to perform local preliminary processing on the collected data. When the sensor detects that a pedestrian runs a red light, the system can directly trigger the local early warning mechanism without waiting for the response of the remote server, thereby reducing the potential risks caused by delays. After the data collection is completed at the perception layer, the data is transmitted to the network layer. This study uses communication protocols such as Wi-Fi, Bluetooth, and

4G to ensure that data can be stably and efficiently transmitted to the terminal server, laying a solid foundation for subsequent data processing. Subsequently, the data enters the data layer, where the system cleans, fuses and models multi-source heterogeneous data, and achieves accurate monitoring and dynamic perception of the pedestrian crossing environment through algorithms such as target detection, target tracking and target prediction. The data layer not only improves the accuracy and timeliness of data processing, but also provides reliable data support for the visualization and interactive functions of the application layer. Finally, as a window for users to interact with the system, the application layer presents the analysis results of the data layer in an intuitive and easy-to-operate way based on the Carla simulation system under the UE engine and the integrated meteorological API interface. The system is designed with functions such as graphical user interface, log report, real-time warning and notification, so that users can grasp the dynamic situation of pedestrian crossings, providing an important basis for intelligent traffic management and autonomous driving decision-making. The entire system architecture is progressive and closely coordinated, from real-time perception of data to intelligent processing, and then to visual interaction, forming a closed-loop feedback mechanism, which lays a solid technical foundation for the in-depth application of virtual twin environment in the field of intelligent transportation.

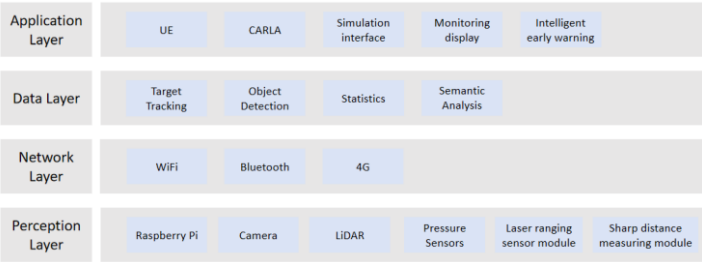


Figure 2. Architecture design diagram of the virtual twin environment system

3.2. Virtual Twin Environment Construction

This study built a twin virtual crosswalk environment based on the integrated architecture of UE and Carla autonomous driving simulation platform. As an open source autonomous driving simulation platform, Carla not only provides a wealth of pedestrian and vehicle model resources, but its locally compiled version can also process OpenStreetMap (OSM) road network data through digital twin tools. The virtual environment built in this study is shown in Figure 3 [18]. The specific process includes: 1) obtaining an .osm format file containing a detailed road structure from the OSM open source map platform; 2) converting it to a Carla-compatible .xodr format using the format conversion tool osm2odr; 3) using the building module library and parameter adjustment system to generate a three-dimensional map containing road, building and vegetation elements to achieve the expected architectural style characteristics; 4) topologically optimizing the converted road network data, and finally building a digital twin scene mapped to the real environment.

3.3. Multi-source Data Collection and Fusion

This paper collects real-time status data of pedestrian crossings by deploying sensors

on site. At the same time, it combines the meteorological API (<https://openweathermap.org>) to obtain real-time environmental data to improve the comprehensiveness of monitoring. Data fusion algorithm and computer vision algorithm are used to integrate sensor data to ensure the accuracy and timeliness of monitoring data.

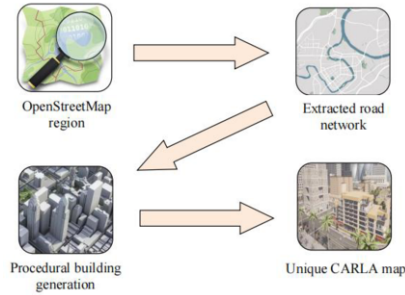


Figure 3. Flowchart of building a virtual environment [18]

3.3.1. Sensor Layout in Pedestrian Crossing Environment

In the monitoring system, the Sharp distance measurement module and the laser distance measurement sensor module are arranged at the gates on both sides of the crosswalk, aiming to detect in real time whether there are pedestrians illegally entering when the red light is on. At the same time, the pressure sensor is installed under the blind path in the waiting area of the crosswalk to sense the waiting status of the blind. When the sensor detects the pressure signal, the system will automatically trigger the voice broadcast function to remind the blind of the current traffic light status and improve travel safety. In addition, the YESKAMO security camera and the RPLiDAR laser radar are used together to monitor the crosswalk and its surrounding environment, providing key data support for subsequent pedestrian and vehicle detection. As an edge control system, the Raspberry Pi is responsible for collecting data and transmitting the data to the server through the network. Table 1 shows the sensors and their models used in the crosswalk environment.

3.3.2. Weather API to get real-time environmental data

In order to obtain the meteorological environment of the study area, this study obtains real-time meteorological data through the OpenWeatherMap API. Its services cover the world and can provide key parameters such as current cloud cover, precipitation, humidity, wind speed, visibility and weather phenomena according to longitude and latitude. The key meteorological parameters are extracted through the JSON data returned by the API to determine the current weather conditions. CARLA controls the weather through the `carla.WeatherParameters` class, which supports the adjustment of cloudiness, precipitation, humidity, wind speed (`wind_intensity`), precipitation (`precipitation_deposits`), visibility (`fog_density`) and other parameters. The mapping relationship in this paper is shown in Table 2. The CARLA parameters are dynamically set according to the weather status returned by the API to form a logical mapping between OpenWeatherMap and CARLA.

According to the experience of the meteorological industry and the meteorological

warning standards of various countries, when the monitoring data exceeds the safety threshold, the system can automatically generate a warning text and send it to the traffic management center or the broadcast notification system next to the crosswalk to assist decision-making. When the total cloud cover exceeds 80%, the weather is already quite gloomy, especially in the case of precipitation or storms, visibility will be further reduced, which will harm traffic. Therefore, the system issues an early warning under this condition. Precipitation warnings are usually based on hourly precipitation. When the precipitation reaches 10 mm/hour, local waterlogging may occur in a short time; when the precipitation exceeds 30 mm/hour, the risk of urban waterlogging increases significantly, and the system issues a preliminary warning at this time. If the precipitation intensity further increases to 50 mm/hour or above, it may cause geological disasters such as flash floods or landslides, posing a serious threat to the safety of urban infrastructure and the lives and property of residents, and the system will issue an emergency warning. When the humidity exceeds 90%, the system issues an early warning. In this environment, haze is easy to form, which not only significantly reduces visibility, but also may cause slippery roads and increase the risk of traffic accidents. In addition, high humidity may promote the growth of mold, posing a threat to the health of people with weak respiratory systems. Wind speed warnings are usually divided according to wind force levels. Generally speaking, when the wind speed reaches 10.8 m/s (level 6 wind), the travel of pedestrians and light vehicles will be affected, and the system will issue a preliminary warning. If the wind speed further rises to 17.2 m/s (level 8 wind) or 24.5 m/s (level 10 wind), it may cause trees to fall, buildings to be damaged, and even endanger key infrastructure such as urban power systems and communication networks. The system will issue an emergency warning. When the depth of accumulated water exceeds 10 cm, street traffic will be seriously affected; if it exceeds 30 cm, many vehicles may find it difficult to drive normally, and there is even a risk of being trapped. In this case, the traffic management department usually needs to activate emergency plans, such as closing some affected roads or organizing emergency drainage to ensure traffic order and pedestrian safety. In terms of visibility warning, when visibility drops to less than 1,000 meters, traffic operations will be affected to a certain extent, and the system will issue a preliminary warning; when visibility further drops to less than 500 meters, flight takeoffs and landings and highway access may be restricted, and an enhanced warning will be issued; if it is less than 200 meters, safe driving cannot be guaranteed, and serious traffic accidents can easily occur, and an emergency warning will be issued at this time.

Table 1. Sensors used in pedestrian crossing environments

Sensor Name	Sensor Model	Function
Raspberry Pi	Pi 5	Control the sensors under the crosswalk, collect sensor data, and upload it to the crosswalk digital twin monitoring system.
Sharp distance measuring module	GP2Y0A710K0F	Distance can be measured contactlessly using infrared rays and PSD (position sensitive detector). They are placed on both sides of the crosswalk and monitor whether pedestrians are running red lights by measuring the distance changes through infrared.
Laser ranging sensor module	AE-VL53L1X	The small laser distance measurement module uses ST Micro's laser distance measurement sensor VL53L1X. It is placed on both sides of the crosswalk and monitors whether there are pedestrians running red lights through the distance changes measured by the laser.
Pressure Sensors	SC902-200kg	The strain gauge is connected to the strain body and outputs a linear voltage according to the load. It is a cantilever load cell. It is placed below the blind crosswalks on both sides of the crosswalk. When a blind person is waiting to cross the crosswalk above, the blind person can be monitored by weight measurement.
YESKAMO Security Camera	YESKAMO	Optical environment monitoring under pedestrian crossings
RPLiDAR	RPLiDAR S1	Point cloud environment monitoring under pedestrian crossing

Table 2. Logical mapping table between OpenWeatherMap and CARLA

Parameter	OpenWeatherMap	CARLA	Mapping logical relationships
cloudiness	clouds.all (0-100%)	cloudiness (0-100)	Direct mapping: Mapping OpenWeatherMap cloud cover percentages to CARLA cloud cover values
precipitation	rain.1h (mm/h)	precipitation (0-100) precipitation_deposits (0-100)	Linear mapping: Assuming the maximum precipitation of 50 mm/h corresponds to 100, the formula is: CARLA precipitation = min(100, rain.1h * 2)
humidity	humidity (0-100%)	wetness (0-100%)	Direct Mapping: Map OpenWeatherMap Cloud Cover Percentages to CARLA Cloud Cover Percentages
wind speed	wind.speed (m/s)	wind_intensity (0-100)	Segment mapping: linear scaling (CARLA = wind speed × 10) for ≤10m/s, fixed to 100 for >10m/s
visibility	visibility (m)	fog_density (0-100)	Reverse mapping: Assuming that the maximum visibility is 10,000 meters, the fog volume is 0. The formula is: fog = 100 - (visibility/100)
weather phenomena	weather[0].id (ID)	sun_altitude_angle	Dynamically adjust according to weather type

3.3.3. Data Fusion and Computer Vision Algorithms

Based on the fusion interaction of cameras, lidar and other road sensors, vehicles and pedestrians can be focused in various weather and lighting environments. The sensor fusion data [19] is as follows (1):

$$D_{receiver} = K_{camera} D_{camera} + K_{lidar} D_{lidar} \quad (1)$$

Where: $D_{receiver}$ is the data received by the data receiving device, K_{camera} and K_{lidar} are the weights corresponding to the sensors, and the weights are adjusted appropriately based on local lighting and visibility conditions. D_{camera} and D_{lidar} are real-time data from the camera and lidar sensor at the crosswalk, respectively.

In the data collection and preprocessing stage, this paper collects RGB image information through the camera, and uses the lidar to obtain point cloud data, and performs time synchronization and spatial calibration on the two to ensure data matching and fusion accuracy. Subsequently, target detection is performed based on the YOLOv8 algorithm to detect pedestrians and vehicles in the image. This paper uses YOLOv8n.pt as the detection model, which can provide good computing performance in resource-constrained scenarios while ensuring high detection accuracy. YOLOv8 uses an end-to-end neural network architecture to extract multi-scale features from the input image, and uses adaptive weighted aggregation (Weighted Box Fusion, WBF) to optimize the merging strategy of the target box, thereby improving detection accuracy. Its specific performance indicators are shown in Table 3:

Table 3. YOLOv8n performance index table

Model	YOLOv8n
Size (pixels)	640
mAP ^{val} 50-95	37.3
Speed CPU ONNX (ms)	80.4
Speed A100 TensorRT ONNX (ms)	0.99
Params (M)	3.2
FLOPs (B)	8.7

In terms of target tracking, this paper uses the ByteTrack algorithm to continuously track detected targets. ByteTrack performs preliminary matching based on high-confidence detection results and incorporates low-confidence detection results into association calculations to improve the stability and robustness of tracking. The algorithm first uses the Kalman Filter for state estimation and combines it with the Hungarian Algorithm for target association, making the tracking process more accurate. At the same time, when facing complex traffic scenarios such as occlusion and target interaction, ByteTrack dynamically adjusts through time series information to effectively reduce target loss and mismatching.

By fusing camera and lidar data, combining YOLOv8 for target detection, and

ByteTrack for target tracking, an efficient and stable solution is provided for pedestrian and vehicle identification in complex traffic environments. It is applied to intelligent traffic management scenarios in the system and provides accurate pre-process data for VLM.

3.4. Heading Vision-Language Models

Intelligent analysis uses the VLM model to parse monitoring data and convert it into natural language descriptions to improve the readability and intuitiveness of the data. Combined with computer vision technology, the use of crosswalks is analyzed and intelligently interpreted. When the system is running, it first collects image data of crosswalks in real time through the Carla simulation platform and on-site sensors, including pedestrian dynamics, vehicle behavior, and traffic signal status, and combines environmental data obtained from the meteorological API, such as weather and lighting conditions. After preprocessing, these multi-source data are input into the VLM model for in-depth analysis. VLM uses its powerful visual encoding capabilities to extract image features, such as the number of pedestrians, location, direction of movement, vehicle speed, parking conditions and other details. More innovatively, VLM not only stays at the level of understanding visual data, but further calls the language decoder to combine these image features with environmental information and generate dynamic monitoring descriptions in natural language. The application of the visual language large model in this article is shown in Figure 4.

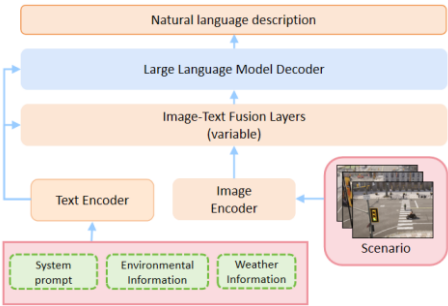


Figure 4. Visual large language model processing flow chart

4. Study Results

4.1. Virtual Twin System Displays Results

The system in this paper builds a digital twin environment based on CARLA, which shows unique advantages in the development of autonomous driving. Figure 5 is the interface diagram of the CARLA twin system. By calling the Python API, the semantic segmentation of the environment and the switching of point cloud views can be realized. Relying on UE, the platform realizes 3D scene rendering close to the real world. It not only accurately simulates vehicle dynamics, complex weather changes and diverse road conditions, but also provides high-precision modeling of multimodal sensors, including the synchronous output of data such as lidar and cameras. This high-

fidelity feature provides a solid foundation for the verification of perception algorithms. Compared with the high-cost model of traditional road testing that relies on physical vehicles, CARLA's virtual simulation environment allows zero-risk reproduction of extreme scenarios, significantly reducing R&D costs and safety hazards. At the same time, the open architecture of the platform gives it a high degree of customization. Users can quickly adjust parameters such as traffic flow and building layout through Python API, and can also directly import real road map data or work with SUMO to achieve macro traffic flow simulation.



Figure 5. CARLA digital twin display page

4.2. VLM Intelligent Early Warning Prompt Results

Intelligent warning uses sensor data and meteorological data to input the visual language big model so that the big model can output intelligent warning prompts. When the system detects that specific meteorological or traffic conditions exceed the safety threshold, the data will be passed as input to the visual language big model. The model combines images, sensor data and historical pattern analysis to automatically generate natural language warning prompts that adapt to the current environment. The output results are shown in Table 4. The model will intelligently output accurate and contextualized warning prompts based on the warning level. These prompts are pushed to pedestrians and traffic management centers in real time through broadcasting to assist decision-making and improve public safety.

4.3. Computer Vision Analysis and VLM Natural Language Generation

Through the target detection algorithm and the target tracking algorithm, the type of target and the target ID can be displayed in the twin system. As shown in Figure 6, the identification of vehicles and pedestrians is marked as "car" and "person" respectively, and the numbers before the target name represent the same target. At the same time, the flow detection line added to the system page is marked with the direction of travel and the flow of traffic in that direction. VLM generates and understands natural language based on the detection algorithm, which prevents the uncertainty of the model and makes the results more accurate.



Figure 6. Twin monitoring system display

Table 4. Intelligent early warning prompt results

Warning Category	Warning level	Intelligent early warning prompts (the big model intelligently provides prompt words that match the scenario)
Cloud cover warning	Above 80%	The sky is overcast, and lighting is insufficient. Please stay alert! Pedestrians should pay extra attention to traffic signals to ensure safe crossing.
Precipitation warning	10mm/h(Preliminary Warning)	Water accumulation may occur in a short time. Please walk carefully and be aware of slippery roads!
	30mm/h(Enhanced Warning)	Increased water accumulation on roads. Please cross cautiously and be mindful of low-lying areas that may flood.
	50mm/h(Emergency Warning)	Heavy downpour may lead to urban flooding and geological hazards. It is advised to avoid going out and follow traffic control instructions.
Humidity warning	90%	High humidity reduces visibility, and roads may be slippery! Pedestrians should walk slowly, watch their step, and wear masks to prevent air pollution effects.
Wind speed warning	10.8m/s(Level 6 Wind, Preliminary Warning)	Strong winds detected. Please secure your belongings and be cautious when crossing streets!
	17.2m/s(Level 8 Wind, Enhanced Warning)	Strong winds may cause tree branches or sunshades to fall. Pedestrians should be cautious and avoid lingering in exposed areas!
	24.5m/s(Level 10 Wind, Emergency Warning)	Severe winds may affect traffic signals and road safety. Pedestrians are advised to pause travel and wait for wind conditions to improve!
Water depth warning	10cm(Preliminary Warning)	Significant road flooding. Please watch your step, be cautious of slippery surfaces, and avoid deep water areas!
	30cm(Emergency Warning)	Some roads are no longer passable. It is advised to take an alternate route or use designated pedestrian crossings!
Visibility warning	1000m(Preliminary Warning)	Visibility is reduced. Pedestrians should stay focused and ensure safe passage!
	500m(Enhanced Warning)	Traffic signals may be affected. Pedestrians should be patient and confirm safety before crossing!
	200m(Emergency Warning)	Extremely low visibility. Avoid going out if possible, and prioritize using subways or indoor passageways!
Pedestrian red light warning	Detected by Distance Sensors	The red light is on. Do not run the red light! Follow traffic rules to ensure safe crossing.
Blind waiting	Detected by Pressure Sensors	A visually impaired person is waiting. Please yield and provide a safe passage.
Traffic warning	High vehicle flow, extended waiting time	Traffic congestion is heavy, and red light wait times may be longer. Please be patient and stay safe!
Crowd warning	High pedestrian density	The pedestrian flow is high. Please maintain a safe distance, walk in an orderly manner, and avoid overcrowding.

The traffic report output by the visual big language model is shown in Figure 7. The content settings include four parts: vehicle conditions, pedestrian conditions, traffic dynamic observation, and potential risk analysis. Through the content of the traffic flow monitoring report, it can be seen that the natural language output of vehicles and pedestrians is completely based on the content displayed in the scene. At the same time, different target values enable traffic managers and other staff to better locate specific vehicles.

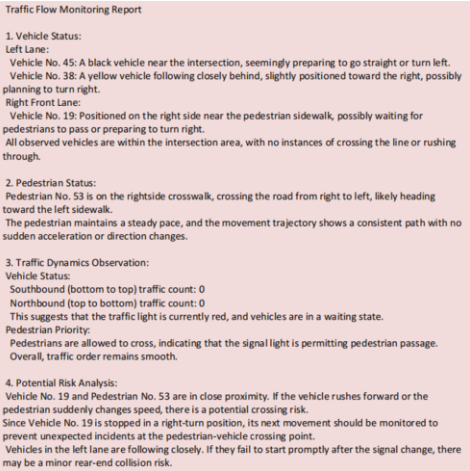


Figure 7. VLM outputs traffic reports

Table 5. Comparison chart of traditional monitoring systems

Comparison Dimensions	Traditional monitoring system	This article twin monitoring system
Data collection method	Mainly rely on field sensors, data is isolated	Fusion of virtual simulation and on-site sensor data, multi-source data fusion
Environmental simulation capabilities	No virtual simulation capabilities, relying only on physical data	Building a virtual environment with UE and Carla
Data processing method	Mainly based on graphical visualization, data analysis is relatively basic	Combined with the visual language model, intelligent data analysis is achieved
Information Expression	Graphical interface display, high threshold for understanding	Automatically generate natural language text to improve information readability
Security early warning capability	Early warning is mainly based on fixed thresholds and is relatively passive	Intelligent threshold warning to improve safety protection capabilities

5. Discussion

The twin monitoring system proposed in this study is compared with the traditional monitoring system. As shown in Table 5, the traditional monitoring system mainly relies on field sensors to obtain data. The data source is single and isolated, and it is difficult to build a complete monitoring system. The system in this paper integrates virtual simulation and sensor data to achieve multi-source data integration, making the monitoring information more comprehensive, and at the same time improving the adaptability and accuracy of the system in different scenarios. In terms of environmental simulation, the traditional monitoring system only relies on physical environmental data and lacks virtual simulation capabilities, so there are obvious limitations in prediction and simulation in complex scenarios. The system in this paper uses UE and Carla to build a virtual environment, so that monitoring can not only be based on real data analysis, but also test the possible situations of different scenarios in advance in the simulation environment, which greatly enhances the applicability and prediction ability of the system and provides more reliable support for risk assessment and decision-making. Intelligent data processing is a highlight of this study. Traditional monitoring systems mainly rely on graphical means to display data. Although they can intuitively present trends, they are insufficient in deep analysis and data mining. The visual language large model in this paper can not only analyze data efficiently, but also automatically generate highly readable analysis conclusions, reduce the difficulty of manual interpretation, and improve the level of intelligent data processing. The optimization of information expression further improves the usability of the system. Traditional monitoring systems usually use graphical interfaces to display data. Although intuitive, they have a high threshold for understanding for non-professionals, which may lead to limited information transmission. In contrast, the twin monitoring system in this paper can automatically generate natural language text and convert complex data into clear and easy-to-understand text reports, making information acquisition more convenient, improving user understanding efficiency, and reducing dependence on professional knowledge. In terms of security early warning capabilities, the system in this paper shows a higher level of intelligence. Traditional monitoring systems usually rely on fixed thresholds for early warning. This method is relatively passive and only triggers an alarm when the monitoring data exceeds the set range, which has a certain lag. The twin monitoring system in this paper adopts an intelligent threshold early warning mechanism, combined with historical data, real-time monitoring information and intelligent algorithms, which can predict potential safety risks and issue early warnings before hidden dangers occur, thereby improving the initiative and accuracy of safety protection.

In the future, there is still a lot of room for optimization of the twin system based on VLM in this paper. We plan to further fine-tune the model to make it closer to the needs of autonomous driving scenarios, and explore the integration of audio data to work with image and text data to achieve true multimodal monitoring. In addition, the system will also introduce human-computer interaction functions, allowing managers to directly communicate with the system through natural language, actively query monitoring status or issue adjustment commands, making the entire monitoring system more intelligent and flexible.

6. Conclusion

This paper proposes a pedestrian crossing environment monitoring system based on virtual-real twins. Using OSM datasets, a twin digital system is built based on UE and Carla simulation platform. At the same time, it combines field sensor data and meteorological API to realize dynamic monitoring of pedestrian crossing status. The system architecture of this paper is constructed by using the hierarchical structure design of the Internet of Things; the YOLOv8 and ByteTrack algorithms are used to perform real-time machine recognition of pedestrian crossing traffic status; the weather conditions obtained by the meteorological API are used to logically map the weather conditions of the twin system, and the data monitored exceeds the safety threshold is warned; the core innovation of this study is to introduce a large visual language model, so that the monitoring data can not only be converted into readable natural language text, but also provide safety warnings and intelligent prompts based on the set thresholds, effectively improving the intelligence, interactivity and readability of the monitoring system. Compared with the traditional monitoring method based on graphical visualization, the virtual twin environment constructed by this system performs better in data processing efficiency and safety warning capabilities, providing a more intelligent, efficient and real-time solution for traffic safety management in the context of autonomous driving. In the future, the system can further optimize the virtual simulation accuracy and VLM's understanding ability to adapt to more complex traffic environment requirements and lay a solid foundation for the development of intelligent transportation systems.

References

- [1] C. Zhang, J. Sprenger, Z. Ni, and C. Berger, "Predicting and analyzing pedestrian crossing behavior at unsignalized crossings," in *2024 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2024, pp. 674–681.
- [2] S. Shen, "Smart pedestrian crossing design by using smart devices to improve pedestrian safety," *Reviews of Adhesion and Adhesives*, vol. 11, pp. 677–690, 2023.
- [3] L. M. Fawzi, S. Y. Ameen, S. M. Alqaraawi, and S. A. Dawwd, "Embedded real-time video surveillance system based on multi-sensor and visual tracking," *Appl. Math. Infor. Sci.*, vol. 12, pp. 345–359, 2018.
- [4] S. Sabbu, R. B. V. Chalavallu, U. Peddireddygar, B. H. Shaik, and P. Badimala, "Real-time vehicle tracking and dynamic signal control through surveillance cameras," in *AIP Conference Proceedings*, AIP Publishing, 2025.
- [5] K. Shaaban, I. Gharraie, E. Sacchi, and I. Kim, "Severity analysis of red-light-running-related crashes using structural equation modeling," *Journal of Transportation Safety & Security*, vol. 13, no. 3, pp. 278–297, 2021.
- [6] T. Driessen, D. Dodou, P. Bazilinskyy, and J. de Winter, "Putting ChatGPT Vision (GPT-4V) to the test: Risk perception in traffic images," 2023.
- [7] C.-Y. Wang, A. Bochkovski, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-

- of-the-art for real-time object detectors,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 7464–7475.
- [8] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, “Pointpillars: Fast encoders for object detection from point clouds,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 12697–12705.
 - [9] M. Dreissig, D. Scheuble, F. Piewak, and J. Boedecker, “Survey on lidar perception in adverse weather conditions,” in *2023 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2023, pp. 1–8.
 - [10] T. Azfar and R. Ke, “Traffic Co-Simulation Framework Empowered by Infrastructure Camera Sensing and Reinforcement Learning,” *arXiv preprint arXiv:2412.03925*, 2024.
 - [11] Y. Wang, Z. Su, S. Guo, M. Dai, T. H. Luan, and Y. Liu, “A survey on digital twins: Architecture, enabling technologies, security and privacy, and future prospects,” *IEEE Internet of Things Journal*, vol. 10, no. 17, pp. 14965–14987, 2023.
 - [12] U. Ullah, M. Usama, Z. Muhammad, and A. Akbar, “AI-enabled low-powered wireless area networks for quality air,” in *Low-Power Wide Area Network for Large Scale Internet of Things*, CRC Press, 2024, pp. 100–141.
 - [13] B. Yang, H. Zhang, M. Du, A. Wang, and K. Xiong, “Urban traffic congestion alleviation system based on millimeter wave radar and improved probabilistic neural network,” *IET Radar, Sonar & Navigation*, vol. 18, no. 2, pp. 327–343, 2024.
 - [14] H. Liu, C. Lin, B. Gong, and D. Wu, “Automatic lane-level intersection map generation using low-channel roadside LiDAR,” *IEEE/CAA Journal of Automatica Sinica*, vol. 10, no. 5, pp. 1209–1222, 2023.
 - [15] F. Yanovsky, “Millimeter-wave radar: principles and applications,” in *Millimeter Wave Technology in Wireless PAN, LAN, and MAN*, Auerbach Publications, 2008, pp. 315–386.
 - [16] K. Castañeda, O. Sánchez, R. F. Herrera, E. Pellicer, and H. Porras, “BIM-based traffic analysis and simulation at road intersection design,” *Automation in Construction*, vol. 131, p. 103911, 2021.
 - [17] S. Robles-Serrano, G. Sanchez-Torres, and J. Branch-Bedoya, “Automatic detection of traffic accidents from video using deep learning techniques,” *Computers*, vol. 10, no. 11, p. 148, 2021.
 - [18] A. Dosovitskiy, “CARLA: An Open Urban Driving Simulator,” *Proceedings of the 1st Annual Conference on Robot Learning*, vol. 78, pp. 1–16, Nov. 2017.
 - [19] H. Zhang, H. Yu, Z. Zheng, S. Yuan, and G. Xiong, “Modeling and application of road intersections based on digital twins,” *Automotive Safety and Energy*, vol. 14, no. 1, pp. 55–61, 2023.